Digital Systems

Mini-Quiz 1

30th January, 2024

Name -	Roll no -

Use white space to solve. You should show all relevant steps to get credit

1. Given the two unsigned binary numbers X = 1111101 and Y = 1100101, perform the following subtractions by using 2's Complement method. [5 marks]
   a) X-Y
   b) Y-X

a) $-Y = 0011010 + 1 = 0011011$

$X + (-Y) = 1111101 + 0011011 = 10011000$ (0011000 = +24)

b) $-X = 0000010 + 1 = 0000011$

$Y + (-X) = 1100101 + 0000011 = 1101000$ (1101000 = −24)

2. The Boolean function F is given below. Write the reduced form for the function F.
   a) $F(A, B) = [(A + B')'.AB]' + [(A + B)(A'B)']'$
   b) $F(A, B, C) = (A + B)(A + B')(A' + C)$ [5 marks]

a) $(A+B')' = A'B$

$(A'B)' = A + B'$

$F = \{(A+B)(A+B')\}' = [A+A]' = A'$

b) $F = A(A'+C) = AC$

Digital Systems

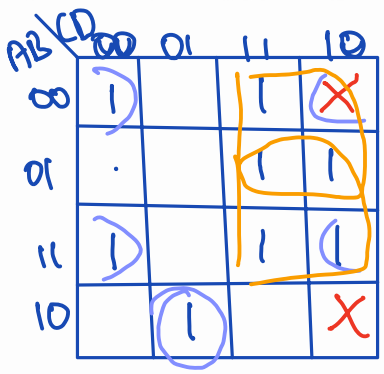
Mini-Quiz 2 (Set A)

6th February, 2024

Name -	Roll no -

Use white space to solve. (Note: all the inputs and their complements are available)

1. Determine the minimized expression for the following below given Boolean function in SOP form using K-map.
   (a)$F(A, B, C, D) = \Sigma m(0, 3, 6, 7, 9, 12, 14, 15) + \Sigma d(2, 10)$



$F = a'b'd' + abd' + ab'c'd + a'c + bc$

(b) Convert the expression obtained in 1.(a) to its NAND gate implementation.